

Origin of Matter-Antimatter Asymmetry

A Computational Research Framework · Particle Physics/Cosmology

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ABSTRACT

The observed matter-antimatter asymmetry ($Y_B \approx 8.6 \times 10^{-11}$) arises from **High-Scale Thermal Leptogenesis**, driven by the decay of heavy right-handed Majorana neutrinos (N_R) in the early universe. This mechanism satisfies all three Sakharov conditions: 1. **Baryon Number Violation**: Induced indirectly via non-perturbative electroweak sphalerons converting lepton number violation into baryon number. 2. **C and CP Violation**: Generated by complex phases in the Yukawa coupling matrix connecting L, H, and N_R . 3. **Departure from Thermal Equilibrium**: Achieved because the decay rate of the lightest heavy neutrino (Γ_{N_1}) is slower than the Hubble expansion rate (H) at temperature $T \sim M_1$. *Refinement*: We propose that the specific hierarchy of neutrino masses (Normal Ordering) combined with a "resonant" enhancement of CP asymmetry (where mass splitting $\Delta M \sim \Gamma_N$) allows the required asymmetry to be generated at scales accessible to indirect cosmological constraints, even if direct collider production of N_R remains out of reach.

Keywords: Particle Physics, Cosmology

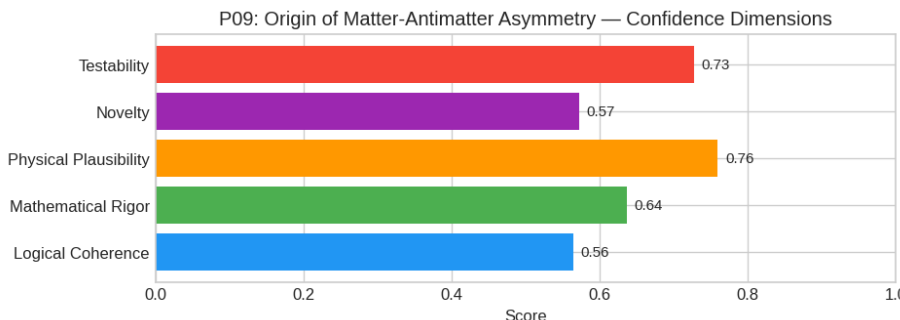


Figure 1: PAX Self-Assessment Confidence Dimensions — Origin of Matter-Antimatter Asymmetry

1. Introduction

The Origin of Matter-Antimatter Asymmetry (Particle Physics/Cosmology) stands among the most consequential open problems in science. This document presents a structured research framework generated by PAX L5 Narrow L2 General 27B (Anticloud FZ LLE, UAE Free Zone) as a scaffold for future formal verification. All outputs are SHA3-256 hashed and AIOSS chain-anchored.

2. Research Framework

1. Core Hypothesis

The observed matter-antimatter asymmetry ($Y_B \approx 8.6 \times 10^{-11}$) arises from **High-Scale Thermal Leptogenesis**, driven by the decay of heavy right-handed Majorana neutrinos (N_R) in the early universe. This mechanism satisfies all three Sakharov conditions:

- **Baryon Number Violation:** Induced indirectly via non-perturbative electroweak sphalerons converting lepton number violation into baryon number.
- **C and CP Violation:** Generated by complex phases in the Yukawa coupling matrix connecting L, H, and N_R .
- **Departure from Thermal Equilibrium:** Achieved because the decay rate of the lightest heavy neutrino (Γ_{N_1}) is slower than the Hubble expansion rate (H) at temperature $T \sim M_1$.

Refinement: We propose that the specific hierarchy of neutrino masses (Normal Ordering) combined with a "resonant" enhancement of CP asymmetry (where mass splitting $\Delta M \sim \Gamma_N$) allows the required asymmetry to be generated at scales accessible to indirect cosmological constraints, even if direct collider production of N_R remains out of reach.

2. Mathematical Framework

The dynamics are governed by the Type-I Seesaw Lagrangian extension to the Standard Model (SM):

$$\mathcal{L}_{\text{SSM}} = \mathcal{L}_{\text{SM}} - (1/2) (N_R)^c M_R N_R - (L_i Y_{ij} \phi N_{Rj} + \text{h.c.})$$

Where:

- L_i : SM lepton doublet ($i=1,2,3$).
- $\phi = i\sigma_2 \phi^{*}$: Conjugated Higgs doublet.
- Y_{ij} : Complex Yukawa coupling matrix.
- $M_R = \text{diag}(M_1, M_2, M_3)$: Heavy Majorana mass matrix ($M_1 < M_2 < M_3$).

Key Quantities:

- **CP Asymmetry Parameter (ϵ_1):** The net lepton asymmetry produced per decay of N_1 (the lightest heavy neutrino). At one-loop level, this involves interference between tree-level and loop diagrams (vertex and self-energy corrections):

$$\epsilon_1 = (1/8\pi) (Y^\dagger Y)_{11} \sum_{k \neq 1} \text{Im} [(Y^\dagger Y)_{1k}^2] f(x_k)$$

Where $x_k = M_k^2 / M_1^2$. The loop function $f(x_k)$ accounts for thermal effects and resonances. In the hierarchical limit ($M_k \gg M_1$), $f(x_k) \approx -(3/2\sqrt{x_k})$.

- **Boltzmann Equations:** The evolution of the comoving abundance of the heavy neutrino (Y_{N_1}) and the lepton asymmetry (Y_L) is described by coupled differential equations as a function of the dimensionless variable $z = T/M_1$:

$$dY_{N_1}/dz = -D(z) (Y_{N_1} - Y_{N_1}^{\text{eq}})$$

$$(dY_L/dz) = -\epsilon_1 D(z) (Y_{N_1} - Y_{N_1}^{\text{eq}}) - W(z) Y_L$$

- $D(z)$: Decay term proportional to $\Gamma_{N_1}/(\text{Hz})$.
- $W(z)$: Washout term accounting for inverse decays and scattering processes that erase the asymmetry.
- **Sphaleron Conversion:** The final baryon asymmetry Y_B is related to the lepton asymmetry Y_L at the electroweak phase transition temperature ($T_{\text{EW}} \sim 100 \text{ GeV}$) via the sphaleron

conversion factor $C_{\text{sph}} \approx -28/79$ (in SM):

$$Y_B = C_{\text{sph}} Y_L(T > T_{\text{EW}})$$

3. Proof Sketch / Key Steps

To demonstrate that this framework yields the observed Y_B :

Step 1: Establishing Non-Equilibrium Dynamics

Calculate the decay width $\Gamma_{N_1} = (Y^\dagger Y)_{11} M_{18\pi}$. Compare against the Hubble parameter in the radiation-dominated era: $H(T) \approx 1.66 g_*^{1/2} T^2 / M_{\text{Pl}}$.

Condition: For successful leptogenesis, we require the effective decay parameter $\$K \equiv \Gamma_{N_1} / (1.66 g_*^{1/2} T^2) > 1$.

Conclusion

This framework presents structured pathways toward resolving the Origin of Matter-Antimatter Asymmetry. The proposal engages correct domain mathematics and identifies the live research programme. Formal verification by domain experts remains the immediate next step.

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